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EE499

Discussion of Findings.

The main finding from this project was that the EMG sensor was able to show a noticeable signal change when the bicep was flexed compared to when it was relaxed. In this setup, the relaxed signal was usually higher, around the upper 800s, while flexing the bicep caused the value to drop, sometimes as low as about 650. Because of this, the analysis used a simple threshold method: if the EMG value dropped below about 830, it was counted as a flexed/contraction state, and if the value stayed above 830, it was counted as a relaxed state. This means the sensor did not need to perfectly measure muscle strength, but it still showed a clear enough difference to detect when the bicep was being used. The biggest issues were noise, wiring, and electrode placement, which made the readings change at times. The budget friendly EMG sensor made the project less accurate than more polished devices like the MyoWare Muscle Sensor, Grove EMG Detector, BITalino, or OpenBCI, which are designed to collect cleaner EMG or biosignal data. Even with those limitations, the project worked as a basic proof of concept because the Arduino and EMG sensor were able to separate relaxed and flexed muscle states using the recorded signal values.